

LESSON 5.2

DETERMINING PROBABILITY



LESSON INTRODUCTION

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

LEARNING TARGETS

- I can determine the probability of an event happening using data.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey and develop a margin of error through the use of simulation.

ESSENTIAL QUESTIONS

- How can the probability of an event, situation, or choice be determined from a data sample?
- How can the probability of an event, situation, or choice be interpreted from a data sample?

LESSON NARRATIVE

This lesson activates prior knowledge of determining probabilities and converting between fractions, decimals, and percentages. Students are given categorical data in different forms, including line plots, bar graphs, frequency tables, and circle graphs, and are asked to calculate and interpret the probability of different events in each situation.

COMPONENT SUMMARY

5.2.1 WARM-UP (~5 MIN.)

Interpreting data presented on a line plot

5.2.2 GUIDED INSTRUCTION (10 MIN.)

Determine probability of events/choices using fractions and percentages using the same data from the 5.2.1 Warm-Up

5.2.3 GUIDED PRACTICE (15 MIN.)

Given two different data sources, practice finding probability from sample data

5.2.4 YOUR TURN! (5-10 MIN.)

Independent practice for students

5.2.5 WRAP-UP (~5 MIN.)

Formative assessment of determining probability from sample data

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have trouble connecting the concept of probability to data samples.
- Students may struggle to rewrite fractions as percentages.

THINGS TO CONSIDER

- This lesson serves as a bridge between the previous one, Unit 5, Lesson 1, and Lessons 3 and 4 from this unit.
- To connect to the prior lesson, let students know that the samples being used are representative of the population, so they can be used to make predictions about the larger population.

LITERACY AND LANGUAGE ROUTINES

Number Talks

The 5.2.1 Warm-Up is an opportunity to allow students to really discuss how they solve each problem along with their strategies. It is an opportunity to not only review what students should already have learned but to also prepare students to connect these concepts to new learning.

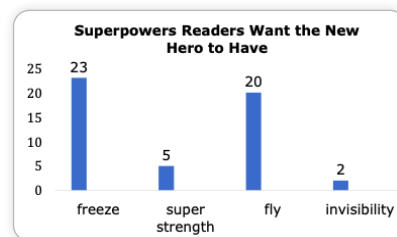
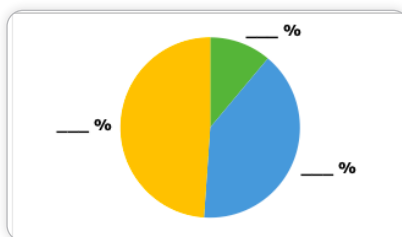
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In the 5.2.3 Guided Practice, students interpret data represented in a variety of ways, including translating from one representation to another. Students also represent probabilities in multiple ways. Consider providing opportunities for students to discuss how they adapt their procedures to new contexts.

DIFFERENTIATE INSTRUCTION

The same data sources are used more than once throughout the 5.2.2 Guided Instruction, 5.2.3 Guided Practice, and 5.2.5 Wrap-Up to allow students to focus more on the concept of probability instead of spending undue time on trying to read data from changing contexts. The data from which students are determining probability are all represented visually in each section as an additional visual aid.



Additional differentiation opportunities are denoted throughout the lesson guide.

5.2.1 WARM-UP: LINE PLOT

TEACHER MOVES

Discuss question 3. This question is strategically placed in the Warm-Up to draw a connection to what students are doing in this lesson: finding a percentage. This should be highlighted before moving on to the next topic since it involves reviewing a skill that students will need to use throughout the lesson.

DIFFERENTIATE INSTRUCTION

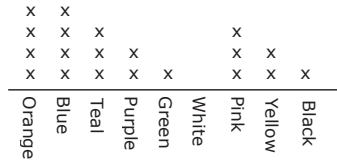
Consider using this as an opportunity for informal observation to inform small groups during 5.2.3 Guided Practice.



5.2.1 Warm-Up: Line Plot

Ms. Field asks the students in her homeroom to make a line plot showing their favorite colors. Each x above a color represents one student's response.

Students' Favorite Colors in Ms. Field's Homeroom



1. How many students are in Ms. Field's homeroom?
There are 20 students in Ms. Field's homeroom.
2. What fraction of students like pink best?
 $\frac{3}{20}$ of the students like pink best.
3. Show how to calculate the percent of students who say green is their favorite color.
 $\frac{1}{20}$ like green the best. To make this a percent, we would multiply the fraction by $\frac{5}{5}$ to create an equivalent fraction out of 100, $\frac{5}{100}$. This means 5% of the students say green is their favorite color.



5.2.2 Guided Instruction: Determine the Probability of an Event

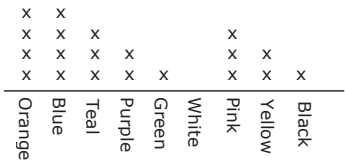
The probability of an event tells us how likely that event is to happen.

1. Match the probabilities with the event described. Write the letter of the event in the space next to the probability of it occurring.

Event	Probability the Event Will Occur
A. A six-sided die lands on a 4	25% B
B. A spinner is divided into four equal sections labeled A, B, C, and D. What is the probability that the spinner lands on a section named with a vowel?	60% C
C. Letters are chosen at random from the word apple. What is probability that a "p" is not chosen?	$16\frac{2}{3}\%$ A

Data can also be used to determine the probability of an event. Here is an example using the line plot from the warm-up.

Students' Favorite Colors in Ms. Field's Homeroom



The line plot can be used to find the probability that a student in Ms. Field's homeroom, chosen at random, picked purple as their favorite color.

2. Find the fraction of students who chose purple as their favorite color.

$$\frac{\text{number of students who chose purple as their favorite color}}{\text{total number of students}} = \frac{4}{20}$$

Recall that probabilities are values between 0 and 1 that can be represented as a fraction, decimal, or percentage

3. Express the probability in each form.

Fraction	Decimal	Percentage
$\frac{1}{10}$	0.1	$\frac{1}{10} = \frac{10}{100} = 10\%$

According to the data, the probability that a student chosen at random from Ms. Field's homeroom chose purple as their favorite color is 0.1, or 10%.

4. What is the probability of a student chosen at random from Ms. Field's homeroom choosing pink as their favorite color?
3 out of 20 or 15% chose pink as their favorite color.
5. What is the probability that a student chosen at random from Ms. Field's homeroom did **not** choose orange as their favorite color?
16 out of 20 students did not choose orange – this reduces to 4 out of 5 or 80%.

5.2.2 GUIDED INSTRUCTION: DETERMINE THE PROBABILITY OF AN EVENT

TEACHER MOVES

This part of the lesson is specifically designed to connect previously learned content about probability to data samples.

ENRICHMENT

Students may struggle to answer Question 5 because the number of students who did NOT choose a color is not explicitly stated. Ask guiding questions like:

- How can we determine who did NOT choose orange as their favorite color?
- How does this differ from finding the percentage of students whose favorite color is purple?
- Are there other ways to phrase this question?

5.2.3 GUIDED PRACTICE: DETERMINE THE PROBABILITY

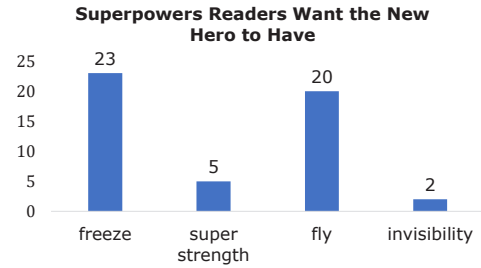
DIFFERENTIATE INSTRUCTION

Consider pulling small groups as needed while students work in order to help students who may have struggled with the 5.2.1 Warm-Up activity.



5.2.3 Guided Practice: Determine the Probability

- Fifty readers of the comic *The Probability Princess* were surveyed on what superpower they would like for the new hero in the series to have. The results are in the bar chart below.



If these 50 people are representative of all the readers of *The Probability Princess*, what is the probability a reader of the series chosen at random wants the new hero to have the power of invisibility?

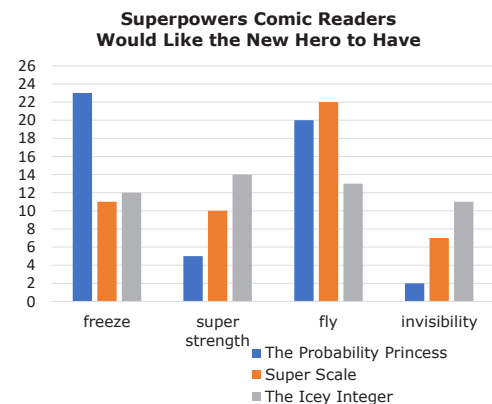
2 out of 50 readers want the power of invisibility – which reduces to 1 out of 25 or 4%.

ENRICHMENT

If you notice students struggling to interpret the graph, ask questions to further scaffold, such as:

- What do you notice about the distribution of the data for the readers of the Icy Integer?
- Are there any trends in the data for the readers of the three series?
- What should the sum of the percentages in each row of the comic book data be?

- Readers of two other comic books series were surveyed as well. Here are the results from the three sets of readers.



- Complete the table to show the probability of each superpower that a randomly selected reader from each category wants the series' new hero to have. Some of the probabilities have already been completed in the table.

	Freeze	Super Strength	Fly	Invisibility
The Probability Princess	$\frac{23}{50} = \frac{46}{100} = 46\%$	$\frac{5}{50} = \frac{10}{100} = 10\%$	$\frac{20}{50} = \frac{40}{100} = 40\%$	$\frac{2}{50} = \frac{4}{100} = 4\%$
Super Scale	$\frac{11}{50} = \frac{22}{100} = 22\%$	$\frac{10}{50} = \frac{20}{100} = 20\%$	$\frac{22}{50} = \frac{44}{100} = 44\%$	$\frac{7}{50} = \frac{14}{100} = 14\%$
The Icy Integer	$\frac{12}{50} = \frac{24}{100} = 24\%$	$\frac{14}{50} = \frac{28}{100} = 28\%$	$\frac{13}{50} = \frac{26}{100} = 26\%$	$\frac{11}{50} = \frac{22}{100} = 22\%$

- b. If a reader is chosen at random, which superpower are they most likely to want the new hero to have?

	Power	Probability
The Probability Princess	Freeze	$\frac{23}{50} = 46\%$
Super Scale	Fly	$\frac{22}{50} = 44\%$
The Icy Integer	Super Strength	$\frac{14}{50} = 28\%$

- c. Using the probabilities, work with your partner to complete the statements.

The writers of *The Probability Princess*

☐ should
☒ should not give the new hero the ability to fly because ...

The readers of this story most wanted to freeze at $\frac{23}{50} = 46\%$, whereas flying had $\frac{20}{50} = 40\%$.

The writers of *Super Scale* ☐ should
☐ should not give

the new hero the ability to fly because ...
This was the most popular choice among its readers – with $\frac{22}{50} = 44\%$.

The writers of *The Icy Integer* ☐ should
☐ should not

give the new hero the ability to fly because ...
The most popular opinion amongst its readers was super strength with $\frac{14}{50} = 28\%$, flying only had $\frac{11}{50} = 22\%$.

- d. Complete the summary.

Readers of three different comic books were surveyed on what superpower they want the new hero in the series to have. If a reader of any of the three series is chosen at random, they are most likely to say the new hero should have the

power to ☐ fly.
☐ freeze. This was also the most

common response in readers of

☐ *Super Scale*.
☐ *The Icy Integer*. However, readers of *The*

Probability Princess are more likely to choose

☐ fly
☒ freeze
☐ invisibility for the new hero's superpower.

The survey shows these readers are nearly evenly distributed among the four choices, making it difficult to predict which power a reader chosen at random from this group would prefer the new hero to have.

5.2.3 GUIDED PRACTICE: DETERMINE THE PROBABILITY (CONTINUED)

TEACHER MOVES

In regard to using probability to make predictions, questions 2c and 2d are purposefully broken down to scaffold the students' understanding of what probabilities they might be able to predict.

QUESTIONING

For students who may be ready for a challenge, consider one of the following questions to extend the content:

- How would the data need to be changed so that the most likely response of readers from any of the three series would be freeze?
- If you were a magazine critic and you had to find a flaw in the data used, what would you write about as the flawed logic of the summary?

5.2.3 GUIDED PRACTICE: DETERMINE THE PROBABILITY (CONTINUED)

TEACHER MOVES

Math Language Routine: Three Reads

For students who need additional support with interpreting word problems, consider framing the Three Reads language routine. Have students just read the information as the first read. On the second read, have students make note or highlight what they notice about the prizes that have been won. For the third read, have students make a prediction about what their circle graph might look like as a self-check for part a.

3. A carnival game has 180 rubber ducks floating in a pool. The person playing the game takes out one duck and looks for a mark on the bottom of it.
- If there is a blue mark, the person wins a small prize.
 - If there is a green mark, the person wins a large prize.
 - If there is no mark, the person does not win a prize.

Rea kept track of how many people won a large prize, a small prize, and no prize.

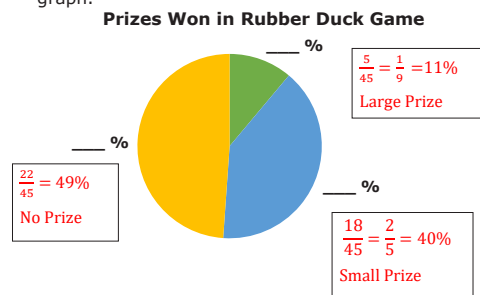
	Number of People
Large Prize	
Small Prize	
No Prize	

ENRICHMENT

If students are struggling to determine the percentage for each section of the graph, ask guiding questions like:

- Based on the data in the frequency table, what should the largest section represent? The smallest?
- How can you use the data from the frequency table to determine the percentages represented by each section of the circle graph?

- a. Rea made a circle graph of the results, but she forgot to label the sections of the graph. Use her data to correctly label the sections of the circle graph.



Use the circle graph to determine the probability of winning a large prize.

- b. Explain how to use the circle graph to determine the probability of winning either a large prize or a small prize.

The graph shows the likelihood of winning a large prize, small prize, or no prize. Read the graph and note blue for small prize, green was for large prize and yellow was for no prize. You can read and interpret the percentage of each. Visually, it is also possible to observe that there is about a 50% chance of not winning any prize (about half the circle) and that winning a large prize is the least likely because it takes up the smallest portion of the circle.



5.2.4 Your Turn!

- Andre picks a block out of a bag 60 times, replacing the block after each pick. He picks a green block 43 of the times.
 - What should Andre estimate for the probability of picking out a green block from the bag?
 $\frac{43}{60} = 71.667\%$
If he were estimating and didn't need an exact answer, he could say this is close to $\frac{40}{60} = \frac{2}{3} = 66\%$.
 - Mai looks in the bag and sees there are 6 blocks. Explain whether Andre should change his probability estimate in light of this new information.
We don't know whether all of the blocks are the same two colors. We know there is a green block and a not green block, but the rest of the blocks could be all green or only one green, so we should keep our experimental probability of $\frac{2}{3}$.
- Before a number cube is sold, it is subjected to rigorous testing. The table below shows how one number cube performed on its three rounds of rolling tests.

	Rolling Test 1	Rolling Test 2	Rolling Test 3
Number of Sixes Rolled	40	42	33
Total Number of Rolls	100	100	100

Based on these results, write a recommendation explaining whether or not this number cube is functioning properly and should be sold.

A number cube should have around $\frac{1}{6}$ odds which is about 17%, all three trials used were about double that percentage. (40%, 42% and 33% respectively). Therefore they should look at the number cube to see whether there are deformities that make it roll a certain way.

5.2.4 YOUR TURN!

ENRICHMENT

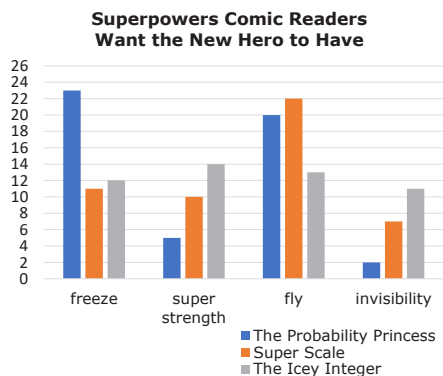
Consider further scaffolding through questioning if you notice students struggling with this section:

- What colors do we know for certain are in the bag? (Green and not-green.)
- Do we know how many of each color are in the bag?
- What are the theoretical probabilities if there is/are only 1 green block? 2 green blocks? 3?
- If there's no way to tell the amount of each color, do we have enough data to change Andre's estimate?



5.2.5 Wrap-Up: Ticket Out the Door

- Use the graph to answer the questions.



- What is the probability that a randomly selected reader of any of the three comics wants the new hero in the series to have the power of invisibility?
 $\frac{20}{150} = 13.33\%$
- How does the probability of a reader of *Super Scale* wanting the new hero to have the ability to fly compare to the probability of a reader of any of the three comic books wanting the new hero to have super strength?
The probability of a reader of Super Scale wanting the new hero to have the ability to fly is $\frac{22}{50} = 44\%$. For any reader, the probability of wanting the hero to have the power of super strength is $\frac{29}{150} = 19.33\%$.

5.2.5 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Because this graph was already used, students are able to focus on the probability for the specified superpower. For students who may struggle analyzing their results, the fill-in-the-blank option provides an example for students by eliminating some of the choices.